FitBrand Co.: Retention, Unit Economics and Capital Reallocation

D2C Acquisition Diagnostics and Action Plan by Channel

1. Executive Summary

FitBrand Co. has successfully scaled customer acquisition across digital channels; however, net margin expansion has decoupled from top-line revenue growth. The primary bottleneck is early-cohort retention: 59.9% of acquired clients churn immediately after order #1. While TikTok Ads generate initial sales volume, they drive severe unit margin dilution due to high churn (21.79% retention rate vs. 54.83% for Ambassador) and higher customer acquisition costs ($15.00 CAC vs. $8.00 Ambassador commission). As a result, an Ambassador-acquired client yields $46.79 in net profit versus $20.31 for TikTok—a +130% efficiency advantage. To maximize net profit without increasing ad spend, this strategic roadmap implements a zero-incremental-budget capital reallocation plan:

1. **Capital Reallocation:** Shift 30% of top-of-funnel TikTok ad budget into the Ambassador commission tier (+15% variable rate) while introducing mandatory High-AOV entry bundles on paid social.
2. **Pre-Depletion Automated Triggers:** Time email/SMS replenishment prompts at N - 3 days prior to product depletion (e.g., Day 27 for Whey Protein), eliminating the observed 5-day purchase delay window.
3. **Post-Purchase Onboarding:** Deploy channel-specific educational sequences to convert impulse TikTok buyers into repeat customers.
4. **Margin-Protected Subscribe & Save:** Implement checkout cross-sell bundles paired with a 10% discount on 2nd deliveries to lock in repeat orders without compressing gross margin below 65%.

2. Business Context and Unit Economics

2.1 The Problem Is Retention, Not Acquisition

Channel-level performance data is conclusive. Ambassador retains 54.83% of customers after their initial purchase; Organic retains 52.94%; Instagram retains 30.48%; while TikTok retains only 21.79%

Data suggests TikTok traffic is driven by impulse buys and low-intent offers, leading to high churn unless paired with a targeted post-purchase re-engagement sequence. Conversely, customers acquired via brand ambassadors or direct search exhibit higher baseline brand trust. With an observed sample size of N = 1,000 customers (~250 per channel), the gap between TikTok's 21.79% retention rate and Ambassador's 54.83% is wide enough to prove a structural difference in purchasing behavior rather than statistical noise.

2.2 CAC vs. LTV per channel

Using the cleaned dataset, three distinct formulas were applied to bridge gross sales and net profit per customer: LTV = total channel sales / number of unique channel customers; Gross Margin = LTV × 65%; Net Margin = Gross Margin − Estimated CAC of the channel. The consolidated result:

Canal	Retention	LTV avg.	Gross margin	Estimated CAC	Net margin / customer
Ambassadors	54.83%	$84.29	$54.79	$8.00	$46.79
Organic	52.94%	$83.38	$54.20	$0.00	$54.20
Instagram	30.48%	$62.42	$40.57	$12.00	$28.57
TikTok	21.79%	$54.32	$35.31	$15.00	$20.31

The analysis reveals a clear channel profitability hierarchy:

- **Organic (Highest Efficiency)**: Leads net profitability with $54.20 USD per customer due to zero customer acquisition cost ($0.00 CAC).
- **Ambassador (Top Paid Channel):** Leads paid/affiliate channels with an LTV of $84.29 USD and a net margin of $46.79 USD per customer.
- **Instagram (Moderate Performance):** Operates at a healthy equilibrium with $62.42 USD LTV and $28.57 USD net margin (deducting $12.00 USD CAC).
- **TikTok (Lowest Return)**: Ranks lowest with an LTV of $54.32 USD. Factoring in its $15.00 USD CAC, net profit drops to $20.31 USD per customer. The unit economic model is fragile: if TikTok's actual CAC exceeds its $35.31 USD gross margin threshold, the channel becomes immediately unviable

Note: Channel CAC benchmarks ($0 Organic, $8 Ambassador, $12 Instagram, $15 TikTok) and the 65% gross margin baseline were established as simulated industry benchmarks for D2C sports nutrition in Latin America. They were specified deliberately to maintain analytical rigor and financial sustainability—they were not extracted directly from the transactional dataset.

3. Data Pipeline Architecture

This dataset is synthetic by design, built in Python using Faker and Pandas to simulate 1,000 unique customers and 1,807 transactions over a 12-month period. The script assigned acquisition channels via a uniform random distribution (~25% initial volume per channel). Subsequently, asymmetric retention probabilities (65% Ambassador/Organic, 40% Instagram, 25% TikTok) were encoded exclusively into the retention probability variable to govern repeat purchases, decoupling initial acquisition from loyalty behavior.

3.1 Category Depletion Cycles

Each product category carries a distinct consumption cycle governing plausible repeat purchase timing:

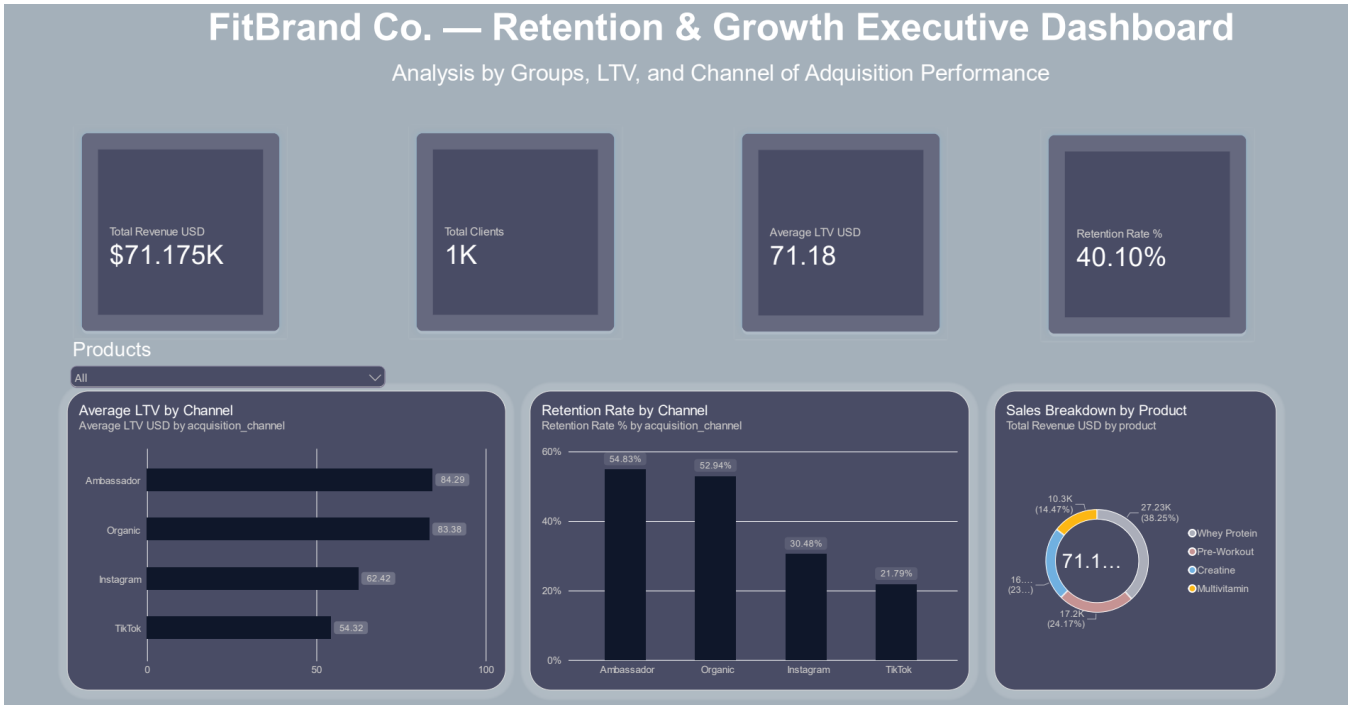
Product	Price	Depletion Cycle
Whey Protein	$55 USD	30 days
Pre-Workout	$40 USD	45 days
Creatine	$35 USD	60 days
Multivitamin	$25 USD	90 days

On the base cycle of each product, a variability of [-5 to +15 days] is introduced before creating the next purchase event. The result of the pipeline is fitbrand_transactions.csv, which serves as raw input data for modeling in SQL and Power BI."

4. Modeling and Analytical Methodology

To evaluate customer lifetime value and repurchase behavior across channels, raw transactional logs (N = 1,807 transactions, 1,000 unique customers) were processed through an analytical pipeline in SQL and Power BI:

- **Channel Retention & Unit Margin Modeling:** Evaluated customer repeat purchase frequency (purchase_number > 1) per acquisition channel. By mapping gross margin (65%) against channel-specific CACs, we established net profit per client across all cohorts.
- **Repurchase Velocity Analysis (SQL LAG Functions):** Calculated exact elapsed days between initial purchase and first replenishment per product category. Comparing empirical repurchase timing against nominal product supply highlighted a ~5-day delay window across core SKUs (e.g., 35 days actual vs. 30 days supply for Whey), pinpointing the exact activation window for automated retargeting.



The Executive Dashboard combines KPI cards (Total Revenue USD, Total Clients, Average LTV USD, Retention Rate %) with channel breakdowns. The visual layout confirms the SQL query output: TikTok's retention bar is noticeably shorter than Ambassador's, and channel average LTV mirrors the same trend, establishing the dashboard as a monthly tracking tool to verify whether budget reallocation is narrowing the gap.

5. Omnichannel Action Plan and Strategy

5.1 Strategic Capital Reallocation & Halo Effect Management

Reallocate 30% of TikTok direct paid ad spend into the Ambassador affiliate commission structure while restructuring paid social entry offers. A customer acquired via Ambassador yields $46.79 USD in net profit compared to TikTok's $20.31 USD (+130% efficiency), driven by lower acquisition costs ($8.00 vs $15.00 CAC) and higher cohort retention (54.83% vs 21.79%).

- **Controlled Reallocation Test:** To account for affiliate channel capacity limits and protect TikTok's top-of-funnel brand awareness ("Halo Effect" feeding Organic search), spend is not eliminated. Instead, low-converting prospecting budgets are shifted to affiliate incentives.
- **Variable Commission Structure:** Elevate code-attributed commissions from 10% to 15% for top-tier creators to increase organic posting frequency without upfront fixed ad costs.
- **Paid Social CAC Absorption:** Restructure TikTok entry offers from single-SKU items to mandatory High-AOV Entry Bundles ($80–$90 USD AOV) to absorb the $15.00 CAC on the first transaction.
- **Channel Maintenance:** Maintain current Instagram spend ($12.00 CAC, $28.57 net margin/client) as a stable mid-funnel volume driver.

5.2 Pre-Depletion Lifecycle Triggers (SQL Repurchase Optimization)

Empirical cohort analysis reveals a delay gap between product consumption and repurchase (e.g., Whey Protein 30-day supply is repurchased at Day 35; Pre-Workout 45-day supply is repurchased at Day 51). Triggering retargeting campaigns after depletion exposes customers to competitor switching. Automated lifecycle triggers are established at pre-depletion activation windows (N - 3 days) to capture repurchase intent prior to stockout:

- **Whey Protein** (30-day supply | 35-day actual repurchase): Trigger at Day 27 (captures 5-day delay).
- **Pre-Workout** (45-day supply | 51-day actual repurchase): Trigger at Day 42 (captures 6.4-day delay).
- **Creatine** (60-day supply | 65-day actual repurchase): Trigger at Day 57 (captures 5-day delay).
- **Multivitamin** (90-day supply | 95-day actual repurchase): Trigger at Day 87 (captures 5-day delay).

5.3 Channel-Specific Post-Purchase Onboarding

Paid social buyers (TikTok/Instagram) exhibit severe early cohort churn (70–78%) due to impulse buying behavior. Deploy an automated 30-day post-purchase education sequence (workout integration, recipe guides, product usage routines) exclusively for paid media buyers to close the product engagement gap and elevate paid social retention closer to Ambassador baselines (~54%).

5.4 Margin-Protected Subscription & Bundling Strategy

To resolve the 59.9% overall cohort churn after order #1 without diluting the 65% gross margin baseline:

- **High-AOV Checkout Bundling:** Display dynamic checkout bundles (e.g., Pre-Workout + Creatine) at checkout for single-SKU entry buyers to maximize initial Order Value (AOV).
- **Margin-Safe Subscribe & Save:** Combine entry bundles with a 10% recurring delivery discount timed around pre-depletion triggers. Expanding AOV on the initial order protects net dollar margins against subscription discount compression.

6. TECHNICAL AUDIT & ANALYTICS PIPELINE

6.1 Data Pipeline Architecture

The underlying dataset simulates N = 1,807 transactions across 1,000 unique customers over a 12-month period, generated via Python (Faker, Pandas). Channel attribution was assigned via uniform distribution (~25% volume/channel), while asymmetric repeat-purchase probabilities were encoded into the retention variable (65% Ambassador/Organic, 40% Instagram, 25% TikTok).

6.2 SQL Modeling & Queries

Transaction logs were ingested into an SQLite relational database (fitbrand.db). Two core queries govern the analytical pipeline:

Retention Rate & Average LTV by Channel

```
SELECT
    acquisition_channel,
    COUNT(DISTINCT client_id) AS total_clients,
    COUNT(DISTINCT CASE WHEN purchase_number > 1 THEN client_id END) AS retained_clients,
    ROUND(
        CAST(COUNT(DISTINCT CASE WHEN purchase_number > 1 THEN client_id END) AS FLOAT) /
        COUNT(DISTINCT client_id) * 100, 2
    ) AS retention_rate_pct,
    ROUND(CAST(SUM(purchase_amount_usd) AS FLOAT) / COUNT(DISTINCT client_id), 2) AS avg_ltv_usd
FROM transactions
GROUP BY acquisition_channel
ORDER BY avg_ltv_usd DESC;
```

Repurchase Velocity via Window Functions (LAG)

```
WITH ordered_purchases AS (
    SELECT
        client_id,
        product,
        purchase_date,
        LAG(purchase_date) OVER (
            PARTITION BY client_id
            ORDER BY purchase_number
        ) AS previous_purchase_date
    FROM transactions
)
SELECT
    product,
    ROUND(AVG(JULIANDAY(purchase_date) - JULIANDAY(previous_purchase_date)), 0) AS avg_days_to_repurchase
FROM ordered_purchases
```

```
WHERE previous_purchase_date IS NOT NULL
GROUP BY product;
```

6.3 Power BI Data Model & DAX Measures

The data model uses transactions as a centralized fact table. Key measures include:

Total Revenue: Total Revenue USD = SUM(transactions[purchase_amount_usd])

Average LTV: Average LTV USD = DIVIDE([Total Revenue USD], [Total Clients], 0)

Retention Rate:

```
Retention Rate % =
 VAR RetainedClients =
    CALCULATE(
       DISTINCTCOUNT(transactions[client_id]),
       transactions[purchase_number] > 1
    )
 RETURN
    DIVIDE(RetainedClients, [Total Clients], 0)
```